UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which data type stores a whole number?
   A. string
   B. float
   C. char
   D. int

2. What is displayed by the following code?

```
for (int i = 1; i <= 4; i++)
    cout << i * 2;
```

   A. 246
   B. 1357
   C. 2468
   D. 1234

3. What is pseudocode?
   A. A C++ header file
   B. Machine code produced by an assembler
   C. A structured English-like description of an algorithm
   D. A type of compiler error

4. How many times does a while loop execute when its condition is false initially?
   A. Two
   B. Zero
   C. One
   D. Until the user stops it

5. Which of the following are selection structures?
   i. if
   ii. switch
   iii. for
   iv. while
   A. ii and iv
   B. iii and iv
   C. i and iii
   D. i and ii

6. Which declaration creates an integer array of five elements?
   A. int a(5);
   B. array a[5];
   C. int a[5];
   D. int a;

7. Which input pattern stops when the user enters -1?
   A. A fixed for loop
   B. A sentinel-controlled loop
   C. A function prototype
   D. A switch statement

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8. What does a compiler do?
   A. Translates the whole source program before execution
   B. Draws flowcharts automatically
   C. Stores only integer values
   D. Executes one source statement at a time

9. Which of the following are valid identifiers?
   i. total_score
   ii. 2total
   iii. _count
   iv. int
   A. iii and iv
   B. ii and iv
   C. i and iii
   D. i and ii

10. Which keyword prevents a variable's value from being changed after initialisation?
   A. const
   B. static
   C. void
   D. break

PART B (60 MARKS)

QUESTION 1

This question tests the program development life cycle.

a) Define source code and object code.

(3 marks)

b) Describe THREE stages of the program development life cycle.

(2 marks)

QUESTION 2

A concert charges the ticket prices shown in Table 1. A negative age is invalid.

Table 1. Concert Ticket Prices

Age Range	Ticket Price (RM)
0 - 12	15.00
13 - 17	25.00
18 and above	40.00

a) Draw a flowchart for the ticket price selection.

(6 marks)

b) Write pseudocode and a complete C++ program for the ticket price selection.

(7 marks)

QUESTION 3

This question tests type conversion and constants.

a) Explain the difference between implicit and explicit type conversion, and give one C++ example of each.

(5 marks)

b) Write a C++ program segment that calculates the average of three integer marks as a double and displays it to two decimal places.

(6 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
    s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(10 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(5 marks)

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QUESTION 5

A sports club records the sport names and medal counts in Table 1.

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

Table 2. Sport and Medal Records

Sport	Medals
Swimming	8
Badminton	12
Athletics	7
Cycling	5
Hockey	9

(8 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(8 marks)

PART C (30 MARKS)

QUESTION 1

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 3. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

QUESTION 2

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

END OF QUESTION PAPER

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